

★ WEEK 13 · PART 1 of 3

Bayesian Inference: Beliefs Updated by Evidence

Part III Opens — Treat Parameters as Distributions, Not Numbers

Two views of probability · Bayes' theorem · posterior, likelihood, prior

Last week's calibration gave single numbers. Bayesian inference gives distributions — a full account of what we believe and how sure we are. It starts with one equation: Bayes' theorem.

1

PART 1 • THE CORE IDEA

Bayes' Theorem and the Update

Two ways to think about probability, Bayes' theorem as the rule for updating belief, the working form posterior proportional to likelihood times prior, and the shift to parameters as distributions.

The Minister's Posthumous Idea — Part 1

Reasoning backward, from evidence to cause

Thomas Bayes, an 18th-century Presbyterian minister, asked a backward question: given what we observe, what should we believe about the cause? His essay on this was published only after his death, in 1763.

The backward question

- Forward: given a cause, how likely is the evidence?
- Backward: given the evidence, how likely is the cause?
- Bayes gave a rule to reverse the direction.

Why it is powerful

- It matches how we actually learn from data.
- It updates a belief in light of new evidence.
- One theorem captures the whole process.

The Minister's Posthumous Idea — Part 2

A long eclipse, then a computational comeback

Laplace developed the idea further, but for much of the 20th century frequentist statistics eclipsed it. Bayesian methods returned only when cheap computing made their once-intractable calculations feasible.

The eclipse

- Frequentist methods dominated 20th-century statistics.
- The Bayesian calculations were simply too hard by hand.
- The idea waited, out of the mainstream, for decades.

The comeback

- Computers made the hard integrals tractable.
- Simulation methods (next week) unlocked real models.
- Bayes is now everywhere in modern data analysis.

From Point Estimates to Distributions

Where Week 12 leads

1

We calibrated to single numbers

Week 12 fit κ , θ , σ as point estimates. But a point hides how UNCERTAIN we are about each.

2

Bayes gives a distribution instead

Rather than one best value, we get a full distribution over the parameter — belief plus uncertainty, together.

3

One rule drives it all

Bayes' theorem updates a prior belief with data into a posterior. This week builds that machinery from the ground up.

Part 1 in one line: Bayes' theorem turns a prior belief plus data into a posterior distribution.

Two Ways to Think About Probability

The frequentist and the Bayesian

There are two coherent views of probability. The frequentist reads it as a long-run frequency; the Bayesian reads it as a degree of belief. The difference shapes what a parameter even IS.

Frequentist

- Probability is a long-run frequency of events.
- A parameter is a fixed, unknown constant.
- Randomness lives in the data, not the parameter.

Bayesian

- Probability is a degree of belief.
- A parameter is uncertain — described by a distribution.
- Belief updates as data arrives.

The Frequentist View

A fixed parameter, random data

The core idea

- The parameter has one true, fixed value.
- We estimate it from a sample of data.
- Repeat the sampling and the estimate varies.

What it gives

- Point estimates and confidence intervals.
- Statements about the PROCEDURE across repetitions.
- No probability attached to the parameter itself.

The Bayesian View

An uncertain parameter, a distribution of belief

The core idea

- The parameter is uncertain, described by a distribution.
- We start with a prior belief about it.
- Data updates that belief into a posterior.

What it gives

- A full distribution over the parameter.
- Direct probability statements about it.
- Uncertainty quantified honestly, not hidden.

Bayes' Theorem: The Update Rule

The single equation behind everything this week

$$P(\theta \mid \text{data}) = \frac{P(\text{data}|\theta) P(\theta)}{P(\text{data})}$$

The pieces

- The posterior: belief about the parameter after data.
- The likelihood: how probable the data is for each value.
- The prior: belief before seeing the data.

The denominator

- The evidence normalizes the result to a distribution.
- It is the probability of the data overall.
- It is also the computationally hard part.

The Four Ingredients

Prior, likelihood, evidence, posterior

What you bring in

- The prior: your belief before the data.
- The likelihood: the model linking parameter to data.
- The evidence: the overall probability of the data.

What comes out

- The posterior: your belief after the data.
- It blends the prior with what the data says.
- Everything Bayesian flows from these four.

Posterior is proportional to Likelihood times Prior

The form you will use constantly

$$\underbrace{P(\theta \mid \text{data})}_{\text{posterior}} \propto \underbrace{P(\text{data} \mid \theta)}_{\text{likelihood}} \times \underbrace{P(\theta)}_{\text{prior}}$$

Why drop the denominator

- The evidence doesn't depend on the parameter.
- So it is just a constant that rescales the result.
- The SHAPE comes from likelihood times prior.

How you use it

- Multiply the prior by the likelihood.
- Normalize at the end so it integrates to one.
- This is the everyday form of Bayes' theorem.

Reading the Working Form

Belief times evidence, rescaled

The multiplication

- Where the prior and likelihood agree, the posterior is high.
- Where either is near zero, the posterior is too.
- The two combine, they don't simply average.

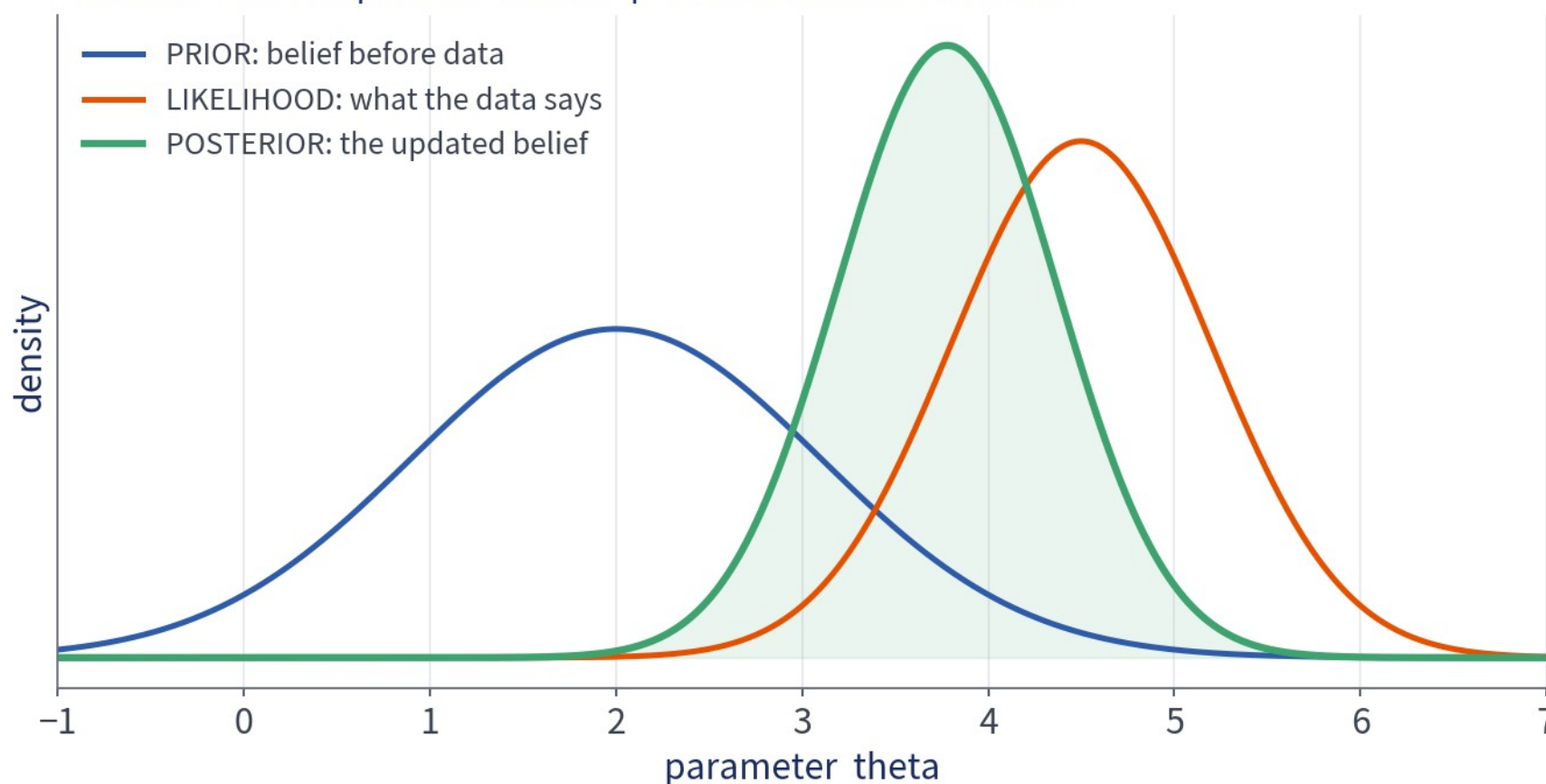
The intuition

- The prior weights values by belief.
- The likelihood weights them by fit to the data.
- The posterior keeps values that satisfy both.

Prior + Data = Posterior

The prior (blue) is belief before data; the likelihood (orange) is what the data says. The posterior (green) sits between them and is tighter than either — the compromise of belief and evidence.

Posterior = the compromise between prior belief and the evidence



The Update as a Compromise

How far the data moves your belief

What sets the balance

- A confident (narrow) prior resists the data.
- A diffuse (wide) prior lets the data dominate.
- More data pulls the posterior toward the likelihood.

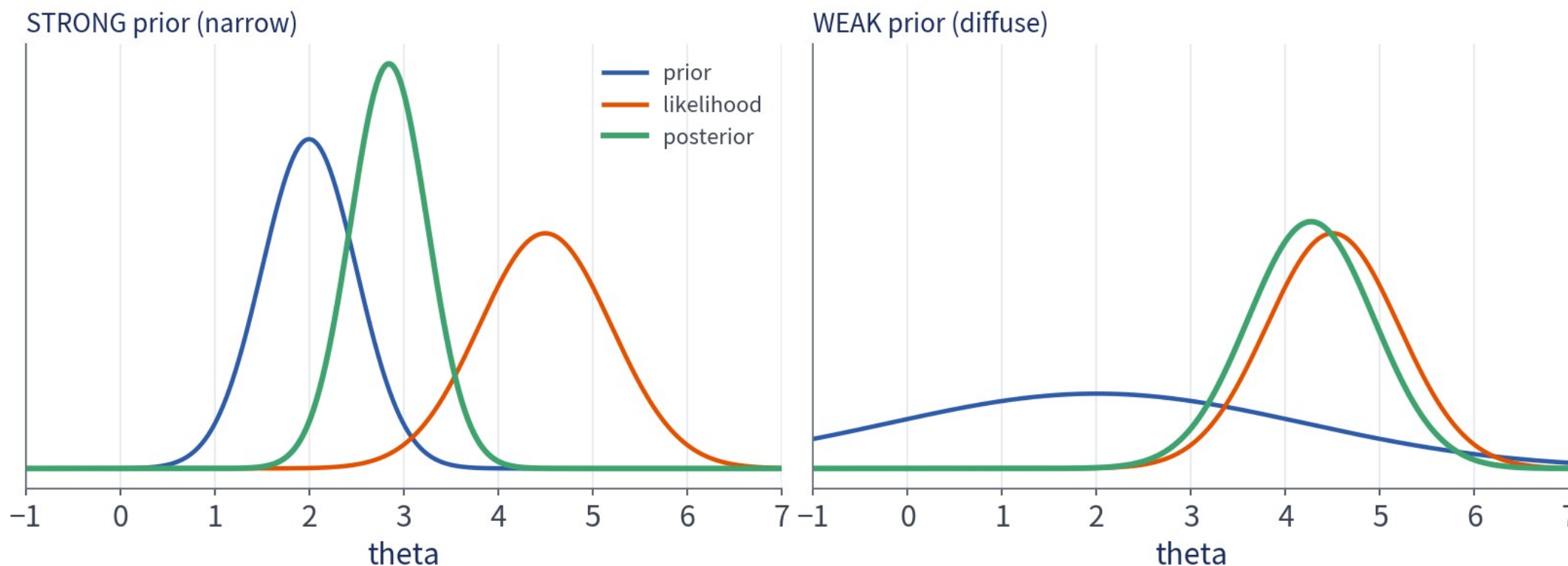
The healthy behavior

- With little data, the prior matters more.
- With lots of data, the prior washes out.
- Belief and evidence trade off sensibly.

Prior Strength Shapes the Posterior

With the same data, a strong (narrow) prior holds the posterior near the prior belief, while a weak (diffuse) prior lets the likelihood lead. The prior's confidence decides how much the data moves you.

A strong prior pulls the posterior toward belief; a weak one lets the data lead



Parameters as Distributions, Not Points

The change in mindset that unlocks everything

The defining move of Bayesian thinking: a parameter is not a single unknown number but a distribution. Instead of 'the value is 0.62', you say 'here is the whole range of plausible values, with their probabilities'.

The old way

- Report one best estimate of the parameter.
- Uncertainty is a separate afterthought.
- The number hides how sure you are.

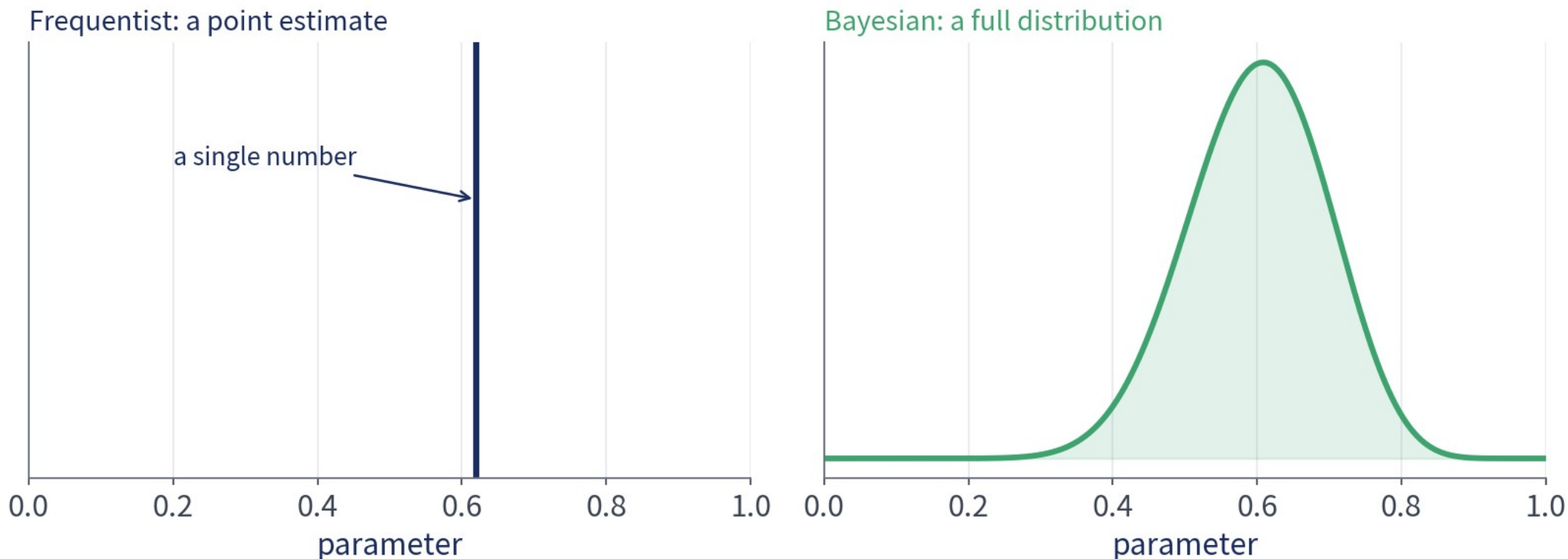
The Bayesian way

- Report a full distribution over the parameter.
- Its spread IS the uncertainty.
- Belief and confidence live in one object.

A Point Versus a Distribution

The frequentist reports a single number; the Bayesian reports a whole distribution. The width of that distribution is the honest measure of how much the data has, and hasn't, pinned the parameter down.

Parameters as distributions, not points



What a Posterior Tells You

More than a single number

It is a full belief

- The most probable value (its peak).
- The uncertainty around it (its spread).
- The probability of any range you care about.

What you can do with it

- Summarize it by a mean or a credible interval.
- Feed it forward into predictions.
- Combine it with more data later.

The Prior: Where Belief Starts

Your knowledge before the data

What it encodes

- What you believed before seeing this data.
- It can be informative or deliberately vague.
- It is a genuine input, chosen with care.

Why it is useful

- It injects real domain knowledge.
- It stabilizes estimates when data is scarce.
- It is also the most debated Bayesian ingredient.

The Likelihood: What the Data Says

The model linking parameter to observations

What it is

- The probability of the data for each parameter value.
- It is the same model a frequentist would use.
- It is where the observations enter.

Its role

- It reweights the prior toward data-consistent values.
- More data makes it sharper and more decisive.
- It is the evidence in Bayes' theorem.

The Evidence: The Normalizer

Why Bayes waited for computers

$$P(\text{data}) = \int P(\text{data} \mid \theta) P(\theta) d\theta \quad (\text{the normalizer - often the hard part})$$

What it is

- The overall probability of the data.
- It integrates the likelihood over all parameter values.
- It rescales the posterior into a proper distribution.

Why it is hard

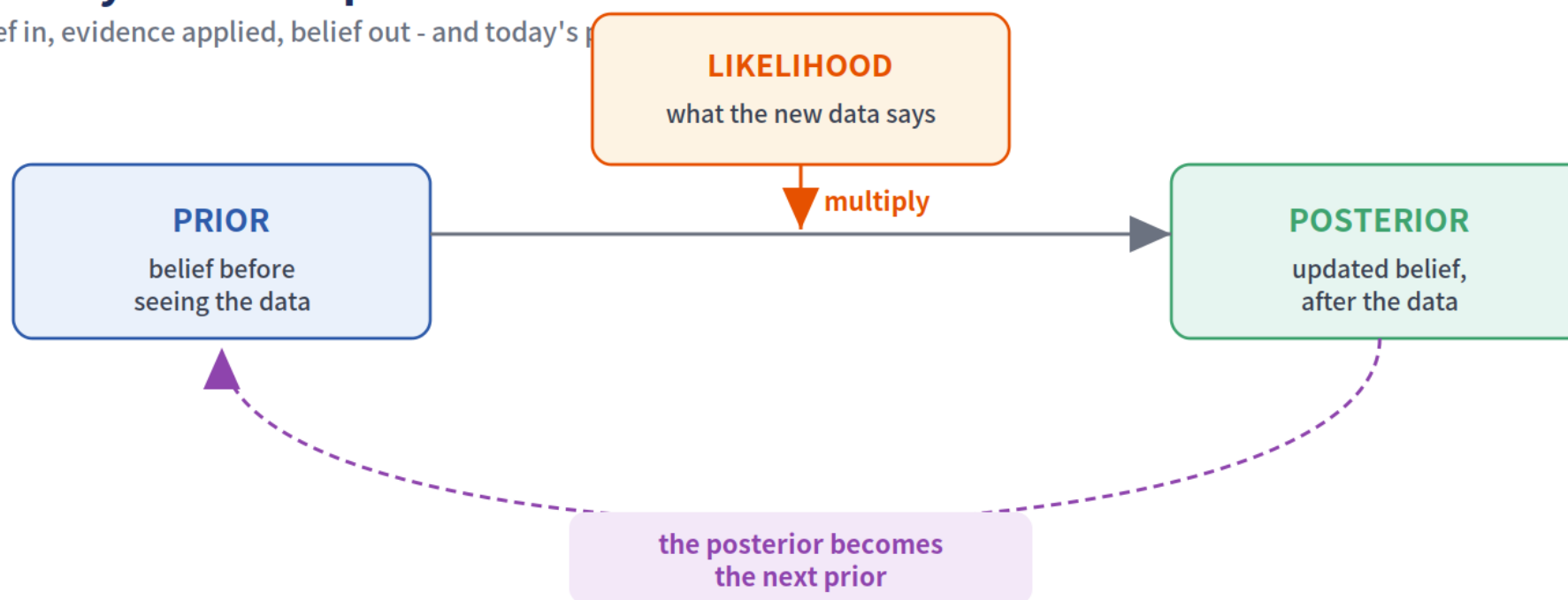
- That integral is usually intractable by hand.
- In many dimensions it is impossible to do exactly.
- This is the problem MCMC solves next week.

The Bayesian Loop

A prior meets new data through the likelihood and becomes a posterior. That posterior then serves as the prior for the next batch of data — learning is an iterative loop, not a one-shot calculation.

The Bayesian Loop

Belief in, evidence applied, belief out - and today's p



Learning is iterative: each batch of evidence updates the belief, and that belief carries forward to the next.

Bayes' Theorem by Hand

Direct Claude Code through one clean update

GOAL Do a single Bayesian update numerically and see the posterior emerge from prior times likelihood.

[INPUT] Prompt · Claude Code

Write and run `bayes_update.py` that:

1. puts a Normal PRIOR on a parameter `theta`;
2. defines a Normal LIKELIHOOD from some data;
3. forms the POSTERIOR on a grid as prior times likelihood, then normalizes it to integrate to 1;
4. plots prior, likelihood, and posterior together, and prints the posterior mean and a 95% interval.

Show the posterior lies between prior and likelihood and is narrower than either.

How to work with it

- Posterior = prior x likelihood, then normalize.
- It sits between the two inputs.
- It is tighter than either alone.
- The grid makes the normalizer explicit.

Deliverable: `bayes_update.py` + a prior-likelihood-posterior plot and a posterior summary.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1 To a Bayesian, what is a parameter?

A distribution, not a fixed number

Q2 In the working form, the posterior is proportional to what?

Likelihood times prior

Q3 Which ingredient is the computationally hard part?

The evidence (normalizer)

All three come from the Bayesian mindset, the working form, and the normalizing integral.

Three Misconceptions to Avoid

Where Bayesian intuition slips

Common myths

- "The prior is just a fudge factor you make up."
- "With enough data, the prior still dominates."
- "The posterior is a single best estimate."

The reality

- The prior encodes real knowledge and can be tested.
- More data washes the prior out — the likelihood wins.
- The posterior is a full distribution, not one number.

Three Things to Carry Forward

Part 1 in one screen

1

Bayes updates belief with evidence

Bayes' theorem turns a prior belief and a likelihood into a posterior. In its working form, the posterior is proportional to likelihood times prior — belief reweighted by the data.

2

Parameters are distributions

Instead of one best number, a Bayesian carries a full distribution over the parameter. Its spread is the uncertainty, stated honestly rather than hidden behind a point estimate.

3

The normalizer is the catch

Computing the evidence — the integral that rescales the posterior — is the hard part. It is exactly why Bayesian methods waited for computers, and it motivates next week's MCMC.

Next — Part 2: the Beta-Binomial, sequential updating, conjugate priors, and the role of the prior.